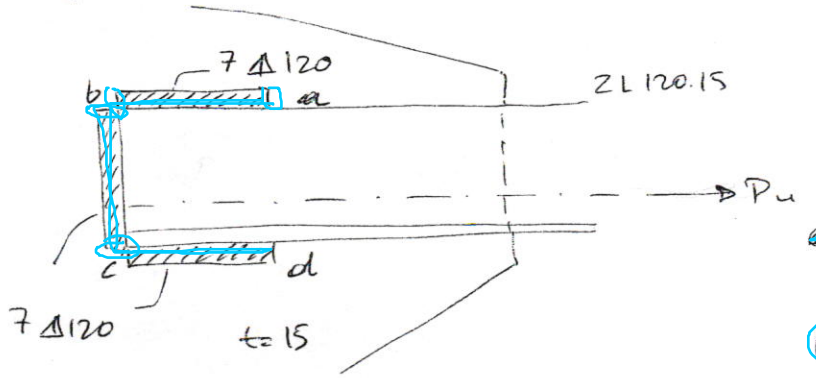
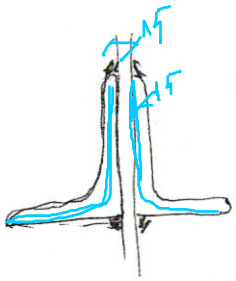
Question 1

A connection detail which is under pure tension effect given in figure. $P_D = 300$; $P_L = 200$ kN. Steel material is S355.





S355
$F_y = 355$ MPa
$F_u = 550$ MPa

E70
$F_u = 70$ ksi
$\cong 480$ MPa

(1 ksi = 6,89 MPa)

L 120.15
$l_x = l_y = 3.51$ cm
$A_g = 33,9$ cm²

Note: welds at a-b, b-c and c-d are applied seperately.

a) Check the maximum stress in connection. (Use a conservative approach)
b) Please re-arrange weld lengths so that makes this connection a simple connection.

Thickness check

$t = 15$ mm
$t = 18$ mm

$3 \leq t \leq 19 \longrightarrow a_{min} = 4.0$ mm

$0,7\,(t - 0,2) = 7$ mm $\quad a_{max} = 7$ mm ✓

Length check

$l = 120 - 2.7 > (6.7 = 42$ mm $/ 40$ mm$)$

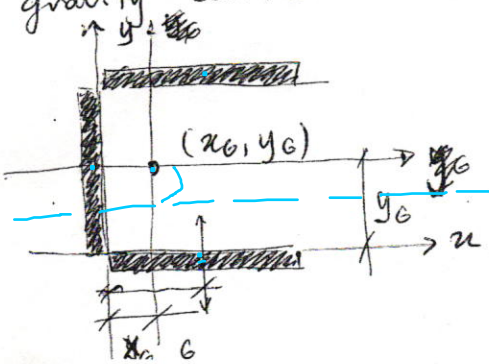
$L = 120 - 2.7 < 150.7 = 1050$ mm

$\rightarrow L_e = L$

$L_e = 120 - 2.7 = 106$ mm

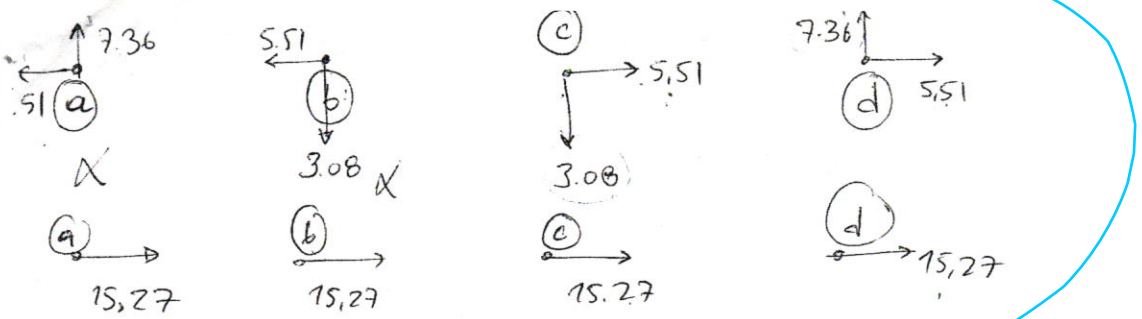
$A_{e_1} = 0,7 \cdot (10,6) = 7.42$ cm²

→ Since the connection is eccentric gravity center should be located.



$y_G = 6$ cm

$x_G = \dfrac{A_{e1} \cdot 6 + A_{e1} \cdot 6 + A_{e1} \cdot 0}{3 A_{e1}} = 4$ cm



f_{mx}, f_{my}

f_{px}, f_{py}

b point

$f_x = 15,27 + 5,51 = 20.08$ kN/cm²

$f_y = 0 + 7,36 = 7.36$ kN/cm²

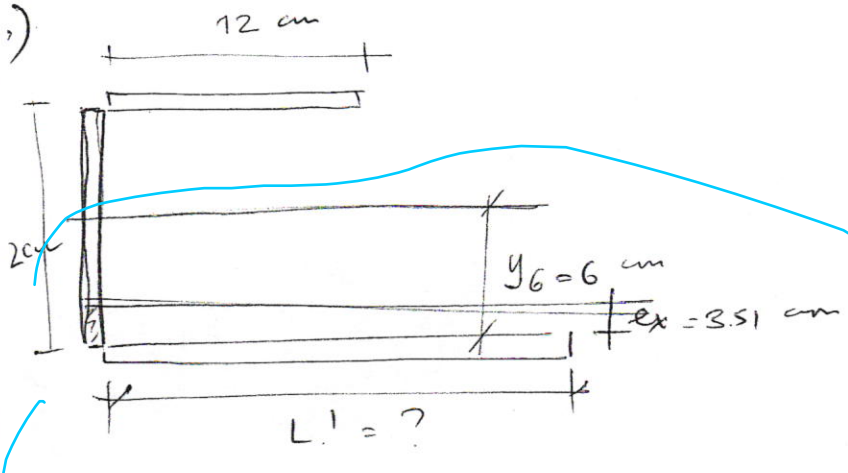
$f_R = \sqrt{f_x^2 + f_y^2} = 22,14$ kN/cm²

$F_E = 70$ ksi $\cong 48,0$ kN/cm²

$F_{nw} = 0,6 \cdot 48 = 28$ kN/cm²

$\phi F_{nw} = 0.75 \cdot 28 = 21$ kN/cm² $< f_R = 22,14$ kN/cm² (Not safe)

c)



$$(L' \cdot a) \cdot e_x + (a \cdot e_x)\frac{e_x}{2} = ((12 - e_x) \cdot a)\frac{12 - e_x}{2} + (12 \cdot a)(12 - e_x)$$

$$(0,7 \cdot L') \cdot 3,51 + (0,7 \cdot 3,51) \cdot \frac{3,51}{2} = ((12 - 3,51) \cdot 0,7)\left(\frac{12 - 3,51}{2}\right) + (12 \cdot 0,7)(12 - 3,51)$$

$$2,457 \cdot L' + 4,312 = 25,228 + 71,316$$

$L' = 37,53$ cm

If bottom weld length would be applied as determined L'; connection would not have eccentricity thus no moment.

$$I_x = \frac{0,7.(10,6)^3}{12} + 2.(7,42.6^2) = 603,72 \ cm^4$$

$$I_y = 2.\left(\frac{0,7.(10,6)^3}{12} + 7,42.2^2\right) + 7,42.4^2 = 317,03 \ cm^4$$

$I_o \cong J \cong I_x + I_y = 608,72 + 317,03 = 920,75 \ cm^4$ → Polar moment of inertia

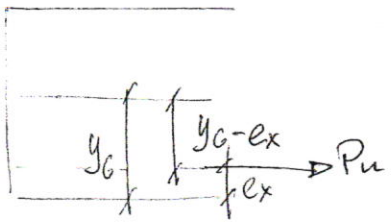
$P_u = 1,2.P_D + 1,6 P_L = 1,2.300 + 1,6.200 = 680 \ kN$

→ Calculation for one-side of the connection, applied force can be divided by half.

$P_u = \frac{680}{2} = 340 \ kN$ (for each side)

→ moment due to eccentricity.

$e = y_G = e_x = 6 - 3,51 \ cm = 2,49$; $M_u = e.P_u = 2,49.340 = 846,60 \ kNcm$



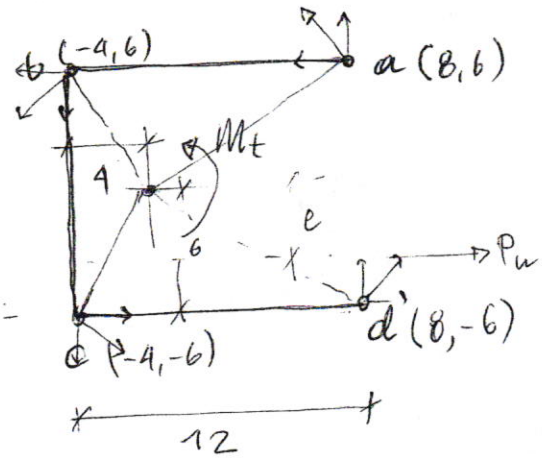
$$f_p = \frac{P}{A_{weld}} \ ; \ f_{px} = \frac{P_x}{A_{weld}} \ ; \ f_{py} = \frac{P_y}{A_{weld}}$$

$$f_m \ (\text{torsional moment}) = \frac{M_t}{J}.r \ ; \ f_{mx} = \frac{M_t}{J}|y| \ ; \ f_{my} = \frac{M_t}{J}|x|$$

x, y: abs. distances

Stress due to tension → $f_{px} = \frac{340}{3.7,42} = 15,27 \ kN/cm^2$; $f_{py} = 0$

Stress due to torsional moment



(a) $f_{mx}^a = \frac{846,6}{920,75}.6 = 5,51 \ kN/cm^2$ $\quad f_{my}^a = \frac{846,6}{920,75}.8 = 7,36 \ kN/cm^2$

(b) $f_{mx}^b = \frac{846,6}{920,75}.6 = 5,51$ $\quad f_{my}^b = \frac{846,6}{920,75}.|-4| = 3,08 \ kN/cm^2$

(c) $f_{mx}^c = \frac{846,6}{920,75}|-6| = 5,51$ $\quad f_{my}^c = \frac{846,6}{920,75}.|-4| = 3,08 \ kN/cm^2$

(d) $f_{mx}^d = \frac{846,6}{920,75}.|-6| = 5,51$ $\quad f_{my}^d = \frac{846,6}{920,75}.8 = 7,36 \ kN/cm^2$